Harun Ardiansyah, S.T., M.S., PhD.
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EDUCATION

August 2021 - December 2025	**Ph.D. in Nuclear, Plasma, and Radiological Engineering** **University of Illinois Urbana-Champaign**, Urbana, IL, USA PhD Dissertation Title: **A Reactor Physics Framework to Detect Anomalies in HTGR Core** PhD. Supervisor: **Prof. Tomasz Kozlowski**
September 2019 - May 2021	**M.S. in Nuclear Engineering and Radiological Sciences** **University of Michigan**, Ann Arbor, MI, USA Research Topic: **Analysis of Steady State and Transient of PBMR-400** Advisor: **Prof. Thomas Downar**
August 2014 - October 2018	**S.T. (Equivalent to B.Eng.) in Nuclear Engineering** **Universitas Gadjah Mada**, Indonesia Final Thesis Title: **Core Design Parametric Study of Integral Pressurized Water Reactor (IPWR) with Mixed Oxide Ceramic Fuel using SRAC Code System** Final Thesis Advisor: **Dr. Alexander Agung**

WORK EXPERIENCE

April 2026 – Present	**Lecturer** Department of Nuclear Engineering, Republic of Indonesia Defense University, Bogor, Indonesia
August 2022 – December 2025	**Graduate Teaching Assistant** • NPRE 455 (Neutron Diffusion and Transport), NPRE 247 (Modeling of Nuclear Energy Systems), NPRE 501 (Fundamentals of Nuclear Engineering)
August 2022 – December 2024	**Graduate Research Assistant** Illinois Microreactor Research, Development, and Demonstration Center (IMRD2C), University of Illinois Urbana-Champaign, Urbana, IL
August 2021 – December 2025	**Graduate Research Assistant** ARTS Group, University of Illinois Urbana-Champaign, Urbana, IL • Performing analysis on neutronic and thermal hydraulic safety of micro-HTGR reactors. • Performing application of neutron noise for advanced reactors safety. • Developing codes for neutron noise in HTGR.
November 2019- May 2021	**Graduate Research Assistant** NURAM Group, University of Michigan, Ann Arbor MI • Performed analysis on steady state and transient of PBMR-400 benchmark.

- Performed validation and verification of HTR-10 with Serpent and AGREE.

August 2016 - May 2018 **Teaching Assistant and Laboratory Assistant**
Universitas Gadjah Mada, Indonesia
- Radiation Detection and Measurement Lab.
- Introduction to Nuclear Technology.

SKILLS/EXPERIENCE
- Programming:
  - Proficient: Python, MATLAB, Git.
  - Familiar: C++, MOOSE.
- Radiation Transport:
  - Monte Carlo Neutron Transport: Serpent, MCNP, OpenMC, SCALE (KENO).
  - Deterministic Neutron Transport: SRAC, PARCS.
- Thermal Hydraulics/CFD:
  - Familiar: RELAP, FLUENT
- Structural Material: ABAQUS

PROFESSIONAL ACTIVITIES
- 2025 Summer School in Open Science + Research Software Engineering by the US Research Software Sustainability Institute (URSSI), August 11-15, 2025, University of Alaska Fairbanks.
- Modeling, Experimentation, and Validation (MeV) Summer School 2024 by Argonne National Laboratory, July 29 – August 9, 2024, Argonne National Laboratory.
- Course on “Core Modeling for Transient” by GRE@T-PIONEER, virtual course, January 2024.
- Course on “Deterministic Modeling of Nuclear Reactor Multi-physics” supported by ENEN2+, December 2023, Chalmers University (Sweden).
- Nuclear Innovation Bootcamp 2022 by Nuclear Innovation Alliances, August 2022, University of Wisconsin-Madison.
- 10th International Conference on High Temperature Reactor Technology (HTR2021), virtually in June 2021. Presented a paper entitled “Evaluation of PBMR-400 Core Design Steady State Condition with Serpent and AGREE”
- Session Chair, 6th International Energy Conference ASTECHNOVA, 2021
- Indonesian Nuclear Society (HIMNI), 2021 – Present
- American Nuclear Society, 2019 - Present

HONORS AND AWARDS
- Mavis Future Faculty Fellowship, 2024 – 2025
- Best Team Project Award, MeV School, 2024
- Barclays G. Jones Graduate Fellowship, Grainger College of Engineering UIUC, 2023 – 2024
- Indonesia Endowment Fund for Education (LPDP) Scholarship, Indonesian Ministry of Finance, 2018

PUBLICATIONS
Peer-Reviewed Journals/Conference Proceedings:
1. **Ardiansyah, H.**, Kozlowski, T., Novel Neutron Noise Unfolding Methods for Anomaly Detection (To be presented in PHYSOR 2026)

2. **Ardiansyah, H.**, & Kozlowski, T. (2026). Review of Detecting Anomalies in Power Reactors Using Neutron Noise Analysis. *Nuclear Technology*, 1–22. https://doi.org/10.1080/00295450.2026.2619208
3. Dwijayanto, R.A.P., **Ardiansyah, H.**, Harto, A.W. (2024). Verification and Geometry Optimization of a One Fluid Molten Salt Reactor (OFMSR) with Fixed Volume. ASME J of Nuclear Rad Sci. 1-20. https://doi.org/10.1115/1.4064465
4. **Ardiansyah, H.**, Oktavian, M. R. (2021). Evaluating the Diffusion Approximation Capability on the Integral Pressurized Water Reactor (IPWR) Core Calculation. Atom Indonesia, 47(2), 85. https://doi.org/10.17146/aij.2021.1013
5. **Ardiansyah, H.**, Seker, V., Downar, T., Skutnik, S., Wieselquist, W. (2021). Evaluation of PBMR-400 Core Design Steady State Condition with Serpent and AGREE. Journal of Physics: Conference Series, 2048(1), 012024. https://doi.org/10.1088/1742-6596/2048/1/012024

Book:
1. **Ardiansyah, H.**, Ekadewi, P., Silalahi, D. F., Gunawan, D., Wahyuni, E., Dipayana, G. F., Hardhi, M., Winofa, N. C., Ramadhan, R. A., Hidayat, T. (2022). Indonesia Post-Pandemic Outlook: Strategy towards Net-Zero Emissions by 2060 from the Renewables and Carbon-Neutral Energy Perspectives. In Penerbit BRIN. https://doi.org/10.55981/brin.562